PLAIN-ENGLISH GUIDE · OCTOBER 2026

Retirement Optimizer

How the app compares your retirement choices, how it scores them, and what to check before you act on a result.

Plans and calculations stay on your computer. If you enable the optional cloud AI summary, selected financial figures are sent to that provider.

This is an open-source planning tool for education and exploration, not personalized financial or tax advice. Results depend on assumptions and may contain errors. Review actual decisions with a CFP® professional and/or CPA.

What the app is trying to do

The app compares ways to fund your spending and leave money for your heirs. It tests each strategy against many possible market outcomes and gives it a **modeled score**: the average amount left after estimated taxes, minus a penalty whenever spending falls below your target. The strategy with the highest score is the recommendation.

1 · Your spending comes first

Each year, every strategy pays your spending and the tax it causes before anything else. A conversion that would leave the year short is cut back. If spending still falls below your target, the gap counts against the score as a penalty.

2 · Then, what your heirs receive

The money left at the end of your plan, after the tax your heirs would owe on traditional retirement-account dollars, stated in today's dollars. This is the number the strategies compete on.

3 · Judged across market scenarios

A market scenario is one possible sequence of yearly investment gains and losses over your retirement. The app draws 300. Strategies are screened in stages: the finalists and the comparison baselines are scored on all 300, and strategies being compared always see the same scenarios.

The score helps compare strategies. It is not a promised account balance. The app does not minimize taxes: a conversion pays tax now to avoid tax later, so the lowest-tax strategy and the best-scoring strategy are usually different ones.

Two numbers you will see, and why they differ

The results page shows both for every strategy. They are measured differently, so they rarely match. The figures below are the example couple’s recommended strategy, in today’s dollars.

Modeled score

$5,816,505

The average after-tax estate across 300 market scenarios, minus any unfunded spending. Returns differ every year in every scenario. **This is what the ranking uses.**

Steady-return value

$5,764,596

The same measure if investments earned exactly the assumed 6% every year. One path, no ups and downs. The year-by-year plan, the charts and the lifetime-tax figures come from this path.

Why the ranking uses the average. A steady 6% cannot show what a bad first decade does to a plan that converted heavily early. The 300 scenarios can, and the strategies that survive them best rank higher.

Why neither is your balance. Both are in today’s dollars at a fixed 2% discount, after the 25% tax heirs would owe on traditional money. Use them to compare strategies, not to predict an account statement.

Two choices it helps you compare

How much to convert to Roth, each year

A Roth conversion moves money from a traditional individual retirement account (IRA) into a Roth IRA. You pay income tax on the amount now; qualified Roth withdrawals, including investment growth, are then tax-free. The question is how much, in which years, and when to stop.

When each of you claims Social Security

If you ask it to, the app tries every eligible claim age from 62 to 70 for each person and scores the ages together with the conversions, because later claiming leaves more low-income years to convert in.

Everything else is your input

Your spending, your investments and their assumed return, your inflation rate, how long to plan for, and your bequest goal. The app does not choose these, and it does not model charitable giving, tax-loss harvesting or Social Security’s earnings test.

The nine strategies it starts from

#	Strategy	What it does each year
1	No conversions	Makes no Roth conversions. Withdrawals for spending and required minimum distributions (RMDs) still occur. Always included; every other strategy is compared with it.
2–6	Fill the 10%, 12%, 22%, 24% or 32% bracket	Converts just enough to bring taxable ordinary income to the top of that federal bracket, after your other income, Social Security and RMDs have used part of it.
7–9	Convert up to 5%, 10% or 20% of the starting IRA balance	Converts one fixed dollar amount, sized once from the balance at the start of the plan. With a $1,000,000 IRA: $50,000, $100,000 or $200,000 a year.

Every amount is a **maximum the strategy asks for**. Your caps, your bracket limit, the Medicare-surcharge setting and the money the year needs for spending can all reduce it. A strategy that runs the money out in a later year is kept and scored with its shortfalls, not removed.

The three percentages are a **limited menu**, not a finding. An amount between them — say $75,000 a year — is not tried, and a custom amount is not offered.

Convert for a few years, then stop

Each converting strategy is first built in full, year by year. Then, for every year it converted in, the app adds a version that **stops after that year**. Each version is a separate strategy, and each is simulated to the last year of your plan.

Version	Year 1	Year 2	Year 3	Year 4	Later years
Stop after year 1	$50,000	$0	$0	$0	$0
Stop after year 2	$50,000	$50,000	$0	$0	$0
Stop after year 3	$50,000	$50,000	$50,000	$0	$0
Every year (the full rule)	$50,000	$50,000	$50,000	$50,000	wherever the rule allows

How many strategies result depends on your inputs: most rules stop converting on their own long before the plan ends, and duplicates are kept once. The four households in the written guide produce 97, 43, 112 and 76 strategies.

The example couple

The rest of this deck follows one household — the test couple used throughout the project's documentation. Every figure is what the app produced for them on 21 October 2026.

Who they are

Both 62, married filing jointly, living in California, planning 30 years from 2026.

What they have

$2,500,000 traditional IRA · $400,000 taxable account (cost basis $300,000) · $50,000 Roth

Income and spending

$30,000 pension · Social Security of $2,800 and $1,600 a month at full retirement age, claimed at 67 · $90,000 a year of after-tax spending

Assumptions behind every figure that follows

6% average annual investment return · 3% inflation · 25% tax on traditional retirement money left to heirs · future amounts discounted at 2% a year · market scenarios vary the yearly return with 12-percentage-point volatility around the 6%.

97 strategies were built for them: no conversions, the eight rules, and 88 stop-year versions. With a $2,500,000 IRA, "Convert up to 5%" asks for $125,000 a year; it converts for 28 years and brings 27 versions. "Convert up to 20%" asks for $500,000 a year; it empties the IRA in four years and brings three.

How it picks one

Step 1

Build every strategy

Each rule is walked year by year with the full tax engine, assuming a steady return. Stop-year versions are added. Duplicates are removed.

Step 2

Every strategy, 100 market scenarios

All 97 are simulated on the first 100 of the 300 market scenarios and ranked by their average score.

Step 3

The leaders, all 300

The three best, plus “no conversions” whatever its rank, are run on the remaining 200 scenarios. Only these fully scored strategies can win.

Step 4

Rank and compare

Highest modeled score wins. Each other finalist is compared with the winner scenario by scenario and labeled: Recommendation ahead, Alternative ahead, Practically equivalent, or Too close to distinguish (slide 11).

When claim ages are searched too

Step 1 runs once for every eligible pair of claim ages — 81 pairs for this couple. Only each pair’s three best strategies by steady-return value, plus “no conversions”, go on to the market scenarios (50 at first, then the five leaders on all 300). Your own claim ages are the exception: every strategy at those ages is carried through, so the comparison baselines on the results page rest on the full 300. For this couple the search chose claiming at 70 and 70 with the same 12%-bracket strategy; 70 and 69 was practically equivalent.

What the couple got

Recommendation

Fill the 12% bracket every year

Convert **$103,000 in 2026**; about $107,000–$123,000 a year through 2030; then $48,000–$58,000 a year from 2031, once Social Security takes some of the bracket. $983,746 over 13 years, last conversion in 2038.

$5,816,505

Modeled score: average after-tax estate in today's dollars across 300 market scenarios, minus unfunded spending (none here)

300 of 300

Spending was funded in all 300 simulated market scenarios, for every finalist, so the score alone decided. A model result, not a guarantee for real life.

Compared with the winner	Modeled score	Behind by	Label
Fill the 12% bracket through 2037, then stop	$5,804,753	$11,752	Recommendation ahead
Fill the 12% bracket through 2036, then stop	$5,792,438	$24,066	Recommendation ahead
No conversions	$5,498,744	$317,761	Recommendation ahead

A second couple, where conversions cannot close the gap

Ages 65 and 63, a $500,000 IRA and nothing else, $60,000 a year of spending, Social Security of $2,000 and $1,200 a month from 67. Until the benefits start, every dollar of spending comes out of the IRA. Same assumptions as the first couple.

2039–2050

The year the money first runs short on the steady-return projection, across all 43 strategies: 2039 for filling the 32% bracket, which converts $299,850 of the $500,000 in its first year; 2050 for no conversions. Each market scenario has its own date.

$4 and $44

How far the three finalists — two tiny 10%-bracket strategies and no conversions — are apart on 300 market scenarios, against a $5,000 threshold. Both alternatives read “Practically equivalent”.

24 vs 1

Of 100 simulated futures, the leading strategies fund spending in every year in about 24. Filling the 32% bracket, scored on the same 300 scenarios for contrast, manages it in fewer than 1 and trails the winner by $288,738.

Small conversions and no conversions produced nearly identical results for this household. Larger conversions performed worse. Changing conversions alone did not resolve the spending gap; the decision that would is about spending or claiming, and the What-if page is where to test it.

Reading the results page, top to bottom

The recommendation

The claim ages, the strategy, this year’s amount, and how clearly it leads its closest alternatives.

Compared with your original choices

Your ages and your best conversions beside the recommendation, measured the same way on the same market scenarios.

Close alternatives

The nearest strategies, ranked as the app ranked them, each with one of the four labels.

The four labels

Recommendation ahead — the market scenarios favor the recommendation.

Alternative ahead — the market scenarios favor the alternative.

Practically equivalent — the two performed similarly under the model’s assumptions. This is evidence of a small difference.

Too close to distinguish — not enough evidence to say which is better. This is a different conclusion from the one above.

When alternatives carry either of the last two, an amber note names them and gives the range of this year’s amounts.

Review this year’s action in the context of the full plan

The first year’s conversion is the figure you can act on now; later years rest increasingly on assumptions and will be re-planned next year with real balances. A recommended claim age several years away still shapes today’s plan, because the conversions before it are sized around it.

$103,000

The example couple’s 2026 conversion. Every finalist agreed on it, so the page shows one number with no range.

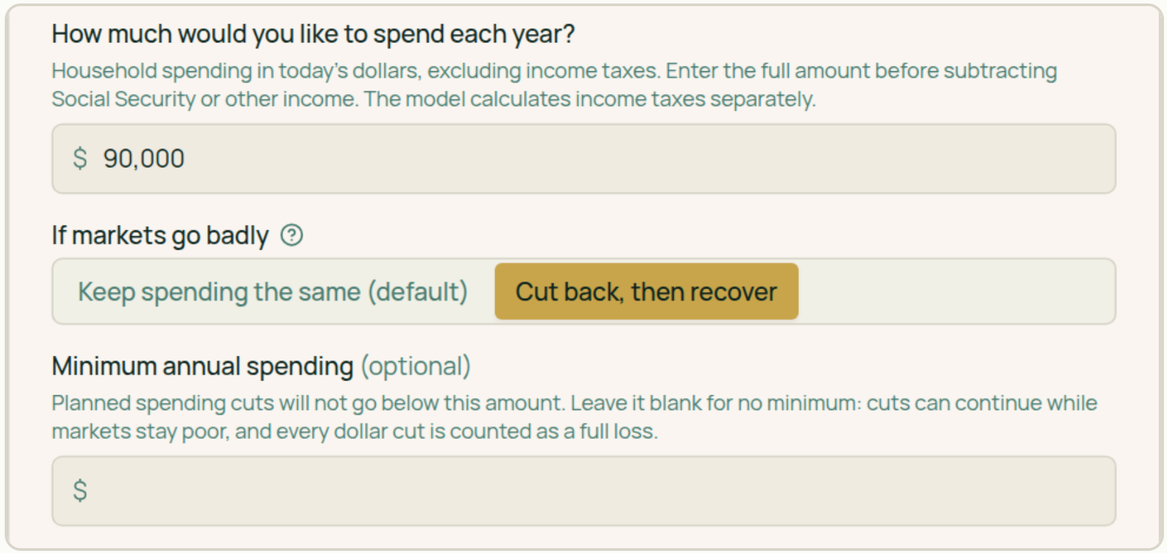
$0

A couple still earning $210,000 of salary: their salaries fill the 10% and 12% brackets, so the recommended strategy converts nothing until 2032. That zero is the result, not a missing strategy.

a range

When alternatives the app cannot separate disagree about this year, it shows the range of their first-year amounts. Compare the two ends on the What-if page before choosing.

Decide how you would spend in a bad market



Step 5 of the plan wizard, *Spending and goals*, with “Cut back, then recover” selected.

Keep spending the same

The default. Your amount, raised for inflation, every year whatever markets do. The strict test of whether the plan lasts.

Cut back, then recover

Spending drops about 10% when a downturn pushes your withdrawal rate 20% above where it started, and is restored when markets recover, never above your target. The optional minimum is a floor the cuts never go below.

Capital Group’s 2026 paper From target date savings to retirement income found that withdrawals set as a share of the balance seldom deplete a portfolio, at the cost of income that swings with the market, while a fixed, inflation-raised amount carries a higher chance of running out late in life. The app’s rule sits between the two. Try both on the What-if page.

Before you act

1. **Check the inputs the result rests on.** The Inputs section of the results page lists every figure the run used; one wrong balance or birth year changes everything after it.

2. **Read the labels, not just the winner.** “Practically equivalent” options performed similarly under the model’s assumptions; “Too close to distinguish” means the model cannot settle it.

3. **Choose how spending reacts to markets.** Step 5 of the plan wizard offers “Keep spending the same” or “Cut back, then recover” (previous slide). Run the plan both ways and compare them on the What-if page.

4. **Test the decision on the What-if page.** Change one thing — spending, the spending rule, a claim age, a death age, the strategy — and see the effect side by side on the same market scenarios.

5. **Take the exports to a professional.** The PDF, Excel and CSV carry the plan, the yearly figures and a Tax Audit sheet a CPA can verify line by line. Review the modeled recommendation with a CFP® professional or CPA before acting.

What the app does not know

About the search

- An amount between the menu's steps — $75,000 when it offers $50,000 and $100,000 — is never tried.
- A later start chosen for its own sake is not a strategy; a late start only arises when a rule finds no room early, or from the conversion-window setting.
- A strategy that trailed on the first 100 scenarios is not reconsidered.
- It finds the best strategy among the ones it builds, not a proof that nothing else would do better.

About the world

- Only the yearly return varies between market scenarios; inflation and lifespan do not, and spending does not unless you chose *Cut back, then recover*.
- Nobody dies in the main projection. "Plan through age" sets only the end year; a death is explored in the lifespan checks and on the What-if page.
- Social Security's earnings test is not calculated; claim ages that overlap a salary before full retirement age are skipped as a modeling restriction.
- No charitable giving, tax-loss harvesting or state-specific rules beyond the tax tables it carries.

THE ONCE-A-YEAR HABIT

Review your plan each year, with real balances.

Update the tax tables when the IRS publishes them, enter the balances you actually have, and re-run so the year ahead rests on facts instead of projections. The plan rolls forward with you.

Written guides: User Guide · How a recommendation is chosen · Verifying results